

Red MMO — Environment Artist Handoff

2026-08-07. Maps: ArtistCanvas + Sandbox_DesertDemoSparkle_T01 (Titan). Merge answer: YES via separate env/gameplay sublevels — never both save one .umap. Ukrainian summaries included at end.

START HERE — Red MMO environmental artist handoff

Generated: 2026-08-07
Ukrainian: START_HERE_UK.md
PDF (EN): RedMMO_Environment_Artist_Handoff.pdf
Merge with gameplay: MERGE_ENV_AND_GAMEPLAY.md · MERGE_ENV_AND_GAMEPLAY_UK.md
Fab / env packs: ENV_FAB_INVENTORY.md
Folder ownership: FOLDER_OWNERSHIP.md

Product name in docs: Red MMO. Paths like Titan.uproject / RedMMOTitan* are technical only.
Gameplay / netcode handoff (separate role): Docs/DeveloperHandoff/ in the same public repo.

60-second path

- 1 Install Unreal Engine 5.8.
- 2 Open the Titan project on the machine that has full Content:
D:\RedMMOTitan\Titan.uproject
(PlanetGen marketplace plugin required for planetary maps.)
 - 1 Open one of the two env / level-design test maps (see below). Do not open both and save blindly.
 - 2 Read MERGE_ENV_AND_GAMEPLAY.md before combining hub graphics with a gameplay map.
 - 3 Run one Unreal Editor at a time on this machine.

The two maps to open

Both live in the Titan monolith (D:\RedMMOTitan), not the clean PPG-only RedMMO project.

#	Soft path	Disk path	Role
1	/Game/RedMMO/Maps/RedPlanetGen_50km_ArtistCanvas	Content/RedMMO/Maps/RedPlanetGen_50km_ArtistCanvas.umap	Planetary environment canvas — continents, atmosphere preview, 50 km planet art
2	/Game/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01	Content/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01.umap	Desert / dressed sandbox for level design, props, lighting experiments

Related (not the primary pair):

Soft path	Project	Note
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/Game/RedMMO/Maps/Tests/RedPlanetGen_50km_FusedPrototype_M07_TropicalTile_R15_V3	Titan	Continent / tropical tile lighting scratch (Region-15)
/Game/RedMMO/Maps/RedMMO_PPG_HomeWorld	Clean RedMMO only: D:\RedMMOTitanWindowsData\Projects\RedMMO\	PPG home world (seed 1337) — planetary presentation; needs PPG plugin

What you own

- Landscape / terrain dressing, foliage, rocks, props, lighting, sky/atmosphere look (within map ownership rules)
- Static meshes and materials under /Game/RedMMO/World/. . . (Quarantine → Approved → Biomes / ManualPOI)
- Hub visuals: architecture dressing, set dressing, fog/post volumes that do not change gameplay rules
- Screenshots from player height, jetpack height, and ship approach

What you must NOT edit

Do not touch	Why
Player pawn / Character Blueprints and C++ (RedPlayerCharacter, Action Trooper setup)	Locomotion and replication
Running animation assets or AnimBP “fixes”	Protected — do not overwrite
Weapon Blueprints, projectile C++, FocalRig aim chain	Gameplay combat
Replication / GameMode / GameState / PlayerController networking	Netcode developer owns
Protected production maps: RedPlanetGen, RedPlanetGen_50km_Test, RedPlanetGen_50km_FusedPrototype	Rollback / acceptance baselines
Marketplace master materials (edit project-owned MI children only)	Pack upgrades break otherwise
Clean RedMMO PPG seed / ProfileV1 generation data unless explicitly tasked	Planetary authority

Engine / plugins for env work

Need	Requirement
UE version	5.8 (D:\UE_5.8 on owner machine)
ArtistCanvas / RedPlanetGen*	PlanetGen (Epic Marketplace for UE 5.8)
Sandbox desert map	Usually opens without PPG; may reference dressed Content packs
Clean PPG home world	Separate project + PPG plugin — only if assigned

First-day checklist

- 1 Open `Titan.uproject` → map `ArtistCanvas` OR `Sandbox_DesertDemoSparkle_T01`.
- 2 Confirm `MapCheck` is clean enough to work (note any existing warnings; do not “fix” gameplay actors).
- 3 Place art only under agreed World folders (see `FOLDER_OWNERSHIP.md`).
- 4 Prefer a new env sublevel for hub dressing rather than saving into the gameplay developer’s map package.
- 5 Export screenshots; list changed folders when you hand work back.

Where this package lives

Copy	Path
Repo docs	<code>Docs/EnvironmentArtistHandoff/</code> (this folder)
Public GitHub	https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff → <code>Docs/EnvironmentArtistHandoff/</code>
Desktop email/PDF pack	<code>C:\Users\user\Desktop\RedMMO_EnvironmentArtist_Handoff\</code>

Environmental / level-design test maps — identity

Status: Identified from on-disk .umap inventory + ProjectKnowledge artist handoff + developer PPG-free map table (2026-08-07).

Primary pair (use these)

Soft path	Project	Disk
/Game/RedMMO/Maps/RedPlanetGen_50km_ArtistCanvas	Titan D:\RedMMOTitan\Titan.uproject	Content/RedMMO/Maps/RedPlanetGen_50km_ArtistCanvas.umap
/Game/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01	Titan same	Content/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01.umap

Why these two

- 1 ArtistCanvas — documented planet-artist canvas (ProjectKnowledge artist_handoff, SAFE8 archive, PlanetArtist README). Built for environmental/planetary authoring review.
- 2 Sandbox_DesertDemoSparkle_T01 — Titan desert sandbox called out for dressed/local env experiments (developer handoff map table). Suitable for level-design prop/lighting tests without mutating protected production planets.

Related maps (do not treat as the primary pair)

Soft path	Project	Note
/Game/RedMMO/Maps/Tests/RedPlanetGen_50km_FusedPrototype_M07_TropicalTile_R15_V3	Titan	Continent / tropical tile lighting scratch (Region-15 PlayerStart)
/Game/RedMMO/Maps/RedPlanetGen_50km_Test	Titan	Protected 50 km desert/ocean baseline — read-only
/Game/RedMMO/Maps/RedPlanetGen	Titan	Production default planet — protected
/Game/RedMMO/Maps/RedMMO_PPG_HomeWorld	Clean RedMMO only	PPG home (seed 1337); not in Titan Content

Protected — do not save over

- /Game/RedMMO/Maps/RedPlanetGen
- /Game/RedMMO/Maps/RedPlanetGen_50km_Test
- /Game/RedMMO/Maps/RedPlanetGen_50km_FusedPrototype

Merging hub / graphics work with the gameplay map

Question: Can an environmental artist build a hub / graphics map while a gameplay developer codes on another map, then merge them without starting over and without overwriting each other?

Answer: YES — if they never both save the same .umap package as the shared ownership file.

Gameplay code in C++ and Blueprint *assets* (Character, GameMode, weapons, AnimBP) can stay completely independent of the environment .umap. Visual hub work should live in env-owned level packages that the persistent level only references.

Recommended Unreal pattern (best)

Use composition, not “one person owns the whole world file.”

```
PersistentLevel (thin shell – rarely edited)
■■■ Sublevel / WP layer: Env_Landscape_Lighting_Foliage    ← artists
■■■ Sublevel / WP layer: Env_HubDressing_Meshes           ← artists
■■■ Sublevel / WP layer: Gameplay_HubLogic                 ← gameplay
    (PlayerStarts, volumes, BP gameplay actors,
     triggers, spawn managers, networked actors)
```

Owner	Owns
Environment artist	Landscape, lighting, fog/post look, foliage, static meshes, set dressing, visual-only volumes
Gameplay developer	GameMode/GameState refs, PlayerStart authority, combat volumes, replicated actors, interactables logic, spawn rules
Tech / lead	Persistent level membership (which sublevels are always loaded), streaming / World Partition rules

World Partition / Data Layers / Level Streaming are all valid implementations of the same idea: separate packages + clear ownership. Prefer World Partition + Data Layers when the hub sits on a large planetary map; prefer classic streaming sublevels for a compact hub sandbox.

What breaks if both edit the same .umap

Reality	Consequence
.umap is a binary Unreal package	Git / Git LFS cannot merge conflicting saves; one writer wins or the file is corrupted / needs full replace
Perforce (better for UE binaries) still needs exclusive checkout	Two people locking/saving the same map = lost work or forced overwrite
Actor GUID / external-actor references	Copy-paste between maps without a plan duplicates or orphans gameplay actors
Autosave + “Save All”	Easy to dirty the wrong map when both are open

Rule: Never have artist and gameplay developer simultaneously authoring and saving HubWorld.umap (or any single shared map). Split first, then compose.

Safe merge options (ranked)

(A) Sublevel / World Partition composition — best

- 1 Create L_Env_HubVisuals (or WP Data Layer Env) from the artist’s dressed map / canvas work.
- 2 Create L_Gameplay_HubLogic from the developer’s gameplay map (or keep logic actors only).
- 3 Persistent level loads both; PIE tests the composition.
- 4 Artists never check in the gameplay sublevel; developers never check in the env sublevel.

Result: Parallel work, no binary map conflict, no restart.

(B) Migrate / copy actors with ownership rules — good if already diverged

- 1 Freeze one side as “source of truth” for structure (usually gameplay PlayerStarts + volumes).
- 2 Migrate env-only actors (StaticMesh, lights, foliage, decals) into the env sublevel via Level → Move into Level / Migrate Asset.
- 3 Do not migrate Character, GameMode overrides, weapon pickups that own replication, or AnimBP assets via map copy.
- 4 Verify collision and PlayerStart once in the composed persistent level.

Result: Recoverable if both already edited different maps; more manual than (A).

(C) One File Per Actor (OFPA) / World Partition external actors — good on WP maps

When World Partition + OFPA is enabled, many actors become individual files under the map’s external actors folder. That reduces whole-map binary conflicts for actors that were externalized, but:

- Persistent map settings, landscape streaming proxies, and non-OFPA actors can still conflict.
- Still require folder ownership and communication.
- Not a substitute for keeping gameplay BP/C++ out of artist packages.

Explicit guarantees

Statement	True?
Merge hub graphics + gameplay map without starting the planet/hub from zero	Yes, via (A) or (B)
Keep gameplay C++ / BP assets while env .umap changes	Yes — code assets are separate packages
Both people save the same .umap and “git merge” later	No — binary conflict; expect overwrite or restore
Artist must re-implement PlayerStart / weapons / netcode after merge	No, if those stayed in gameplay-owned assets/sublevel

Practical workflow for Red MMO

- 1 Gameplay developer works on fundamentals / hub logic (e.g. ThirdPerson / gameplay sublevel / listen-server). Does not dress the ArtistCanvas as the long-term hub visual file.
- 2 Environment artist works on:
 - /Game/RedMMO/Maps/RedPlanetGen_50km_ArtistCanvas for planetary look, or
 - /Game/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01 for desert/LD experiments, or
 - a new L_Env_* sublevel duplicated from either — preferred for hub delivery.
- 1 Lead (or either role with agreement) creates a thin persistent hub that streams both.
- 2 Integration day: load composition, MapCheck, PIE, fix only cross-level references (streaming volumes, soft paths).
- 3 Protected maps (RedPlanetGen, RedPlanetGen_50km_Test, RedPlanetGen_50km_FusedPrototype) stay read-only unless a dedicated mutation gate says otherwise.

Checklist before “we’re merging”

- [] Env work is in env-owned map/sublevel package(s) only
- [] Gameplay actors / PlayerStarts / volumes are in gameplay-owned package(s) only
- [] Persistent level only adds streaming/WP references (minimal dirtying)
- [] No simultaneous save of the same .umap
- [] Running anim, weapons, pawn, replication untouched by art pass

- [] Screenshots of composed PIE from ground + approach

Short answer for stakeholders

Yes, you can merge. Use separate env and gameplay levels under one persistent shell. Do not both edit one binary map. Gameplay code does not need to be rebuilt from scratch when the hub art lands.

Folder ownership conventions (environment)

Aligned with docs/WORLD_AUTHORING_WORKFLOW.md. Artists place and promote content here; gameplay developers do not dump logic Blueprints into these folders.

Content layout

```
/Game/RedMMO/World/Quarantine/<PackName>/      ← new marketplace imports (review only)
/Game/RedMMO/World/Approved/<PackName>/          ← passed pivot/scale/collision/perf
/Game/RedMMO/World/Biomes/<BiomeName>/Palettes/
/Game/RedMMO/World/ManualPOI/<RegionId>/<POIName>/
```

Folder	Who writes	Rule
World/Quarantine/*	Env artist / import pass	Never drag straight into production maps
World/Approved/*	Env after review	Source for placement
World/Biomes/*/Palettes	Env	Data Assets (URedWorldAssetPalette)
World/ManualPOI/*	Env	Hand-authored colonies / landmarks
Maps/RedPlanetGen_50km_ArtistCanvas	Env (canvas)	Do not put networked gameplay actors here long-term
Maps/Sandbox_DesertDemoSparkle_T01	Env / LD experiments	Prefer spawning a new L_Env_* delivery sublevel from it
Maps/RedPlanetGen (+ 50km Test / FusedPrototype)	Protected	Read-only without explicit gate
Character / Weapons / GameMode packages	Gameplay	Artists do not edit

Map package naming (delivery)

Prefer names that make ownership obvious:

```
/Game/RedMMO/Maps/Hubs/L_Hub_Persistent
/Game/RedMMO/Maps/Hubs/L_Hub_Env_Visuals
/Game/RedMMO/Maps/Hubs/L_Hub_Gameplay_Logic
```

Or World Partition Data Layers: DL_Env, DL_Gameplay.

Marketplace masters

- Do not edit Fab/Marketplace master materials or meshes in place.
- Create project-owned Material Instances under /Game/RedMMO/World/... or /Game/RedMMO/Environment/....
- Keep vendor pack folders intact for updates.

Outliner hygiene

- Group env actors under folders: Env/Landscape, Env/Foliage, Env/Lighting, Env/Dressing.
- Group gameplay under Gameplay/Spawns, Gameplay/Volumes, Gameplay/Interactables.
- Add Red Manual Placement Protection on hand-authored POI roots when using the planetary workflow (see world authoring guide).

Fab / Marketplace inventory — environment focus

Subset of Docs/DeveloperHandoff/FAB_MARKETPLACE_INVENTORY.md for environmental / level-design work.

Presence checked on owner machine 2026-08-07.

Legend: Titan = D:\RedMMOTitan\Content · Clean = D:\RedMMOTitanWindowsData\Projects\RedMMO\Content

Plugins required for env maps

Name	Needed for	Notes
PlanetGen (UE 5.8 Marketplace)	ArtistCanvas, RedPlanetGen*	Install via Epic Launcher into UE 5.8
PPG	Only clean RedMMO_PPG_HomeWorld	Not required for Titan ArtistCanvas / desert sandbox
Nwiro / UAIP	Optional editor tooling	Not required for art delivery

Environment content packs

Pack	On Titan	On Clean RedMMO	Typical use
SoStylized	Yes	Yes	Landscape layers, stylized foliage/rocks language
Stylized Desert Oasis	Yes	Yes	Desert oasis water MI, palms, desert dressing
StylizedRocksPack_01	—	Yes	Grass chunks / rock meshes used in PPG stylized binding
TropicalAlienWorld	Yes	—	Cliffs, rock formations, alien tropical silhouettes
AlienJungle	Yes	Yes	Obsidian pillars/rocks, jungle alien props
Alien_Grass_Pack	Yes	Yes	Ground cover candidates (review before approve)
Alien_Plants_Pack	Yes	Yes	Plant props (quarantine → approve)
SpaceColony	Yes	—	Sci-fi architecture dressing
AsteroidSpaceport	Yes	—	Port / hard-surface set dressing
Cloudz_HI5	Yes	—	Cloud volumes (atmosphere dressing; respect scale rules)
Vefects	Yes	—	VFX support (use sparingly in env maps)

Full gameplay/plugin checklist (FocalRig, Steam, etc.) stays in the developer Fab inventory — env artists usually ignore those.

Review before placing on a live map

Score packs (mood-board fit, variation, pivots/LOD/Nanite, editability, cost) per docs/WORLD_AUTHORING_WORKFLOW.md.

Import → world/Quarantine → approve → palette entry → place only approved meshes into ManualPOI / biome folders.

Size / transfer note

Full content/ is tens of GB and is not entirely in git. Artists need a drive sync or shared Content drop for the packs above. The public handoff repo carries docs + fundamentals crumbs, not every marketplace pack.

START HERE — Red MMO передача художнику середовища (UK)

Дата: 2026-08-07

English: START_HERE.md

PDF (EN): RedMMO_Environment_Artist_Handoff.pdf

Злиття з геймплеєм: MERGE_ENV_AND_GAMEPLAY_UK.md · повна EN MERGE_ENV_AND_GAMEPLAY.md

Fab / паки: ENV_FAB_INVENTORY.md

Папки: FOLDER_OWNERSHIP.md

Назва продукту: Red MMO. Шляхи Titan.uproject / RedMMOTitan* — лише технічні.

Окрема роль (геймплей / мережа): Docs/DeveloperHandoff/.

Шлях на 60 секунд

- 1 Встановіть Unreal Engine 5.8.
- 2 Відкрийте Titan: D:\RedMMOTitan\Titan.uproject (для планетарних карт потрібен PlanetGen).
- 3 Відкрийте одну з двох тестових карт (нижче).
- 4 Прочитайте MERGE_ENV_AND_GAMEPLAY_UK.md перед злиттям хаба з геймплей-картою.
- 5 Лише один Unreal Editor одночасно.

Дві карти для роботи

Обидві в проекті Titan (D:\RedMMOTitan).

#	Soft path	Роль
1	/Game/RedMMO/Maps/RedPlanetGen_50km_ArtistCanvas	Планетарний canvas середовища (50 км)
2	/Game/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01	Пустельний sandbox для level design / props / light

Пов'язані (не основна пара): TropicalTile R15 V3 (Titan Tests); RedMMO_PPG_HomeWorld лише в clean RedMMO + PPG.

Не чіпати

- Player pawn / Character BP і C++
- Running animation
- Зброя, снаряди, aim chain
- Реплікація / GameMode / netcode
- Захищені карти: RedPlanetGen, RedPlanetGen_50km_Test, RedPlanetGen_50km_FusedPrototype
- Майстер-матеріали marketplace (лише project-owned MI)

Пакет

- GitHub: <https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff> → Docs/EnvironmentArtistHandoff/
- Desktop: C:\Users\user\Desktop\RedMMO_EnvironmentArtist_Handoff\

Злиття хаба / графіки з геймплей-картою (коротко, UK)

Питання: Чи можна паралельно робити env/хаб і геймплей, а потім злити без старту з нуля і без перезапису?

Відповідь: ТАК — якщо обидві сторони не зберігають один і той самий файл .umap.

Код геймплею (C++ / Blueprint-ассети) живе окремо від карти середовища.

Рекомендований патерн

```
PersistentLevel (■■■■■ ■■■■■■■■)
■■■ Env_* — landscape, light, foliage, meshes ← ■■■■■■■■
■■■ Gameplay_* — PlayerStart, volumes, BP logic ← ■■■■■■■■■■
```

Level Streaming / World Partition / Data Layers — різні реалізації тієї ж ідеї.

Чому не можна «просто змерджити» одну карту

.umap — бінарний пакет. Git / Git LFS не зливає конфлікти карт. Perforce потребує exclusive checkout. Двоє зберігають одну карту → хтось втрачає роботу.

Варіанти (рейтинг)

	Метод	Коли
A	Окремі sublevel / WP (найкраще)	Нова робота хаба
B	Міграція акторів за правилами ownership	Вже є дві роз'їхані карти
C	OFPA / external actors (WP)	Великі WP-карти; не замінює ownership

Гарантії

- Злити hub graphics + gameplay без переробки планети з нуля — так (A/B).
- Залишити C++/BP геймплею при зміні env .umap — так.
- Обидва зберігають один .umap і «змерджать у git» — ні.

Повна EN-версія: MERGE_ENV_AND_GAMEPLAY.md.